

Performance evaluation of deep transfer learning on multi-class identification of common weed species in cotton production systems

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ABSTRACT

Precision weed management offers a promising solution for sustainable cropping systems through the use of chemical-reduced/non-chemical robotic weeding techniques, which apply suitable control tactics to individual weeds or small clusters. Therefore, accurate identification of weed species plays a crucial role in such systems to enable precise, individualized weed treatment. Despite recent progress, the development of a robust weed identification and localization system in the presence of unstructured field environments remains a serious challenge, requiring supervised modeling using large volumes of annotated data. This paper makes a first comprehensive evaluation of deep transfer learning (DTL) for identifying common weed species specific to cotton (*Gossypium hirsutum* L.) production systems in southern United States (U.S.). A new dataset for weed identification was created, consisting of 5187 color images of 15 weed classes collected under natural light conditions and at varied weed growth stages, in cotton fields (primarily in Mississippi and North Carolina) during the 2020 and 2021 growth seasons. We evaluated 35 state-of-the-art deep learning models through transfer learning with repeated holdout validations and established an extensive benchmark for the considered weed identification task. DTL achieved high classification accuracy of F1 scores exceeding 95%, requiring reasonably short training time (less than 2.5 h) across models. ResNeXt101 achieved the best overall F1-score of $98.93 \pm 0.34\%$, whereas 10 out of the 35 models achieved F1 scores near or above 98.0%. However, the performance on minority weed classes with few training samples was less satisfactory for models trained with a conventional, unweighted cross entropy loss function. To address this issue, a weighted cross entropy loss function was adopted, which achieved substantially improved accuracies for minority weed classes (e.g., the F1-scores for Xception and MnasNet on the Spurred Anoda weed increased from 48% to 90% and 50% to 82%, respectively). Furthermore, a deep learning-based cosine similarity metric was employed to analyze the similarity among weed classes, assisting in the interpretation of classifications. Both the codes (<https://github.com/Derekabc/CottonWeeds>) for model benchmarking and the weed dataset (<https://www.kaggle.com/yuzhenlu/cottonweedid15>) of this study are made publicly available, which expect to be a valuable resource for future research on weed identification and beyond.

1. Introduction

Weeds are critical threats to crop production; potential crop yield loss due to weeds is estimated at 43% on a global scale (Oerke, 2006). In cotton production, poor weed management can lead to yield loss of up to 90% (Manalil et al., 2017). Traditionally weed control is performed through tillage and weeding machinery or by hand weeding. With the advent of *trans*-genetic, glyphosate-tolerant crops since 1996, over 90% of the U.S. farmlands for field crops such as cotton, are planted with

herbicide-resistant seeds (USDA-ERS, 2021). The weed control has become predominantly reliant on herbicide application (Duke, 2015; Pandey et al., 2021). Intensive, blanket, broadcast application of herbicides, however, has adverse environmental impacts and facilitates the evolution of herbicide-resistant weeds (e.g., Palmer Amaranth, Waterhemp), which in turn substantially increases management costs (Gould et al., 2018).

Precision weed management (PWM) has emerged as a promising solution for sustainable, effective weed control, which incorporates

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sensors, computer systems and robotics into cropping systems (Young et al., 2014). By recognizing biological attributes of different weeds, PWM enables precise and minimum necessary treatments according to site-specific demand and targeting individual weeds or a small cluster (Gerhards and Christensen, 2003), which can lead to significant reductions in the consumption of herbicides and other resources. For instance, a robotic weeder can spray a particular type or volume of herbicide or use a mechanical weeder or lasers to treat specific weed species, avoiding unnecessary application to crops, bare soil or plant residuals (Barnes et al., 2021). Therefore, successful implementation of truly integrated, precise weed control strategies relies on accurate identification, localization and monitoring of weeds. Machine vision and robotic technologies for automated weed control have recently been demonstrated in certain specialty crops (Fennimore and Cutulle, 2019). However, commercial-scale applicability to row crops such as cotton in varying growing conditions has yet to be evaluated or demonstrated. Lack of a robust machine vision system capable of weed recognition with accuracy exceeding 95% in unstructured field conditions has been identified as one of the most critical technological bottlenecks towards full realization of automated weeding (Westwood et al., 2018). The key to addressing this bottleneck would lie in the development of image analysis and modeling algorithms of high and robust performance.

Image analysis methods based on the extraction of color and texture features, followed by thresholding or supervised modeling, are widely used for weeds classification and detection (Hamuda et al., 2016; Wang et al., 2019). A variety of color indices that accentuate plant greenness have been proposed for separating weeds from soil backgrounds in color or spectral images (Lu et al., 2022; Meyer and Neto, 2008; Woebbecke et al., 1995). The color indices that are developed from empirical observations are however not robust enough in dealing with images acquired under variable field lighting conditions (Hamuda et al., 2016; Lu et al., 2022). In Bawden et al. (2017), texture features including local binary pattern and covariance features were used to perform weed classification, and the extracted features that were applied to a robotic platform, achieved an accuracy of 92.3% on the dataset containing 40 images of 6 weed species. Local shape and edge orientation features were used in Ahmad et al. (2018) for discriminating monocot and dicot weeds, which achieved an overall accuracy of 98.4% based on AdaBoost with Naïve Bayes. In Bakhshipour and Jafari (2018), Fourier descriptors and invariant moments were extracted and fed into support vector machine for classifying four common weeds in sugarbeet fields, resulting in an accuracy of 93.3% accuracy. Despite promising results, the aforementioned image feature-based approaches require engineering hand-crafted features for given weed detection/classification tasks, which may not adapt satisfactorily to a more diverse set of imaging conditions.

Recently, data-driven methods such as deep learning (DL), e.g., convolutional neural networks (CNNs), have been researched for weed classification and detection (Hasan et al., 2021). CNNs are able to capture spatial and temporal dependencies of images through the use of shared-weight filters and can be trained end-to-end without explicit feature extraction (O'Shea and Nash, 2015), empowering neural networks to adaptively discover the underlying class-specific patterns and the most discriminative features. In Dyrmann et al. (2016), a CNN model, which was trained on a dataset containing 10,413 images with 22 plant species at early growth stage, achieved a classification accuracy of up to 98%. A graph-based DL architecture with multi-scale graph representations was developed in Hu et al. (2020) for weed classification, achieving an accuracy of 98.1% on the DeepWeeds dataset (Olsen et al., 2019). While successful, training such DL models from scratch are very time-consuming and resource-intensive, requiring high-performance computing hardware and large-scale, high-quality annotated image datasets, which may not be readily available.

Transfer learning, a methodology that aims at transferring the knowledge across domains, can greatly reduce the training time and the dependence on massive training data by reusing already trained models

for new problems (Zhuang et al., 2020). Therefore, deep transfer learning (DTL, i.e., transferring DL models) only involves fine-tuning model parameters using new datasets in the target domain. DTL has been investigated for weed identification. In Olsen et al. (2019), two pretrained DL models were fine-tuned and tested on the DeepWeeds dataset, achieving average accuracies above 95%. In Espejo-Garcia et al. (2020a), the authors found that fine-tuning DL models on agricultural datasets helped reduce training epochs while improving model accuracies. They fine-tuned four DL models on the Plant Seedlings Dataset (Giselsson et al., 2017) and the Early Crop Weeds Dataset (Espejo-Garcia et al., 2020b) and improved the classification accuracy by 0.51% to 1.89%. In Suh et al. (2018), six pretrained DL models were adopted to classify sugarbeet and volunteer potato images and achieved the accuracy of up to 98.7%. In Ahmad et al. (2021), three pre-trained DL models were used for weed classification, achieving 98.8% accuracy in classifying images of four weed species collected in corn and soybean fields. However, these studies experimented, in addition to a very few weed species, only a handful of DL models. Given active developments in DL model architectures (Khan et al., 2020), it would be greatly beneficial to the research community to evaluate a broad range of state-of-the-art DL models on multi-class weed identification, so as to facilitate informed selection of high-performance models in terms of accuracy, training time, model complexity, and inference speed.

Despite transfer learning strategies, having large volumes of annotated image data is highly desirable for powering DL models in visual categorization tasks (Sun et al., 2017). Currently, the dearth of such datasets remains a crucial hurdle for exploiting the potential of DL and advancing machine vision systems for precision agriculture (Lu and Young, 2020; Mylonas et al., 2021). In weed identification and detection, to achieve high accuracy and robustness requires a dataset that provides adequate representation of important weed species and accounts for the variations associated with environmental factors (e.g., soil types and characteristics, field light, shadows) as well as growth-stage-related morphological or physiological variations. Lu and Young (2020) reviewed 15 publicly available weed image datasets dedicated to weed control, such as DeepWeeds (Olsen et al., 2019), Early crop weed dataset (Espejo-Garcia et al., 2020b), Open Plant Phenotyping Database (Madsen et al., 2020), among some others. Most of these datasets contain a small number (<10) of weed species, with images acquired from a single growth season in geographically similar field sites. No image datasets of weeds specific to cotton production systems have been published so far.

In this paper, we present a new weed dataset collected in cotton fields in multiple southern U.S. states over the two consecutive growth seasons of 2020 and 2021. We establish a comprehensive benchmark of a large set of DL architectures for weed classification on the new dataset. This research is expected to provide a valuable resource for future research on developing machine vision systems for cotton weed control and beyond. The contributions of this paper are highlighted as follows:

1. The presentation of a unique, diverse weed dataset¹ consisting of 5187 images of 15 weed classes specific to the U.S. cotton production systems.
2. A comprehensive evaluation and benchmark of 35 state-of-the-art DL models² through transfer learning for multi-class weed identification.
3. A novel DL-based cosine similarity metric for assisting in the interpretation of DL output and a weighted loss function for improving classification accuracies for minority weed classes.

¹ <https://www.kaggle.com/yuzhenlu/cottonweedid15>.

² <https://github.com/Derekabc/CottonWeeds>.

2. Materials and methods

2.1. Cotton weed dataset

Color or RGB (Red-Green-Blue) images of weed plants were collected from cotton fields using either smartphones or hand-held digital cameras. For the sake of image diversity, following the recommendations in (Lu and Young, 2020), images were captured from different view angles, under natural field light conditions, at varying stages of weed growth, at different locations across the U.S. cotton belt states (primarily in North Carolina and Mississippi). Regular visits to cotton fields were conducted throughout June to August in the growing seasons of 2020 and 2021 for weed image collection. In 2020, images were mainly acquired in the cotton fields of North Carolina State University research stations, including Central Crops Research Station (Clayton, NC), Upper Coastal Plain Research Station (Rocky Mount, NC) and Cherry Research Farm (Goldsboro, NC). In 2021, more weed images were acquired in cotton fields of the R. R. Foil Plant Science Research Center (Starkville, MS) and Black Belt Experiment Station (Brooksville, MS) of Mississippi State University. To facilitate the development of a diverse, large-scale dataset, weed scientists from multiple institutions were invited to participate in the image collection effort. A Google form³ was created and shared for uploading weed images and associated metadata (e.g., weed species, field sites, weather conditions).

The acquired images were first annotated for weed species by weed experts during image submission through the Google form, and the received images were then annotated by trained personnel, and the final annotations were examined again by experts to ensure annotation quality. The images with multiple classes of weeds were cropped so that each resultant image contained a single class of weeds. The weed classes were here defined in terms of common names of weed plants. At the time of writing, the entire dataset contains more than 10,000 images for over 50 weed species, which will be documented in detail in a future study. In this study, the weed dataset used for benchmarking DL models consists of a total of 5187 images for 15 common weed classes. The image number for each weed class is shown in Fig. 1. It should be noted that all weed classes, except Morningglory, correspond to single weed species. The images of different Morningglory species (e.g., Ivy Morningglory, Pitted Morningglory, Entireleaf Morningglory and Tall Morningglory) were grouped together as a single weed class, because of their similarity in weed management and the simplicity of image annotation. Overall, the weed classes including Morningglory, Carpetweed, Palmer Amaranth, Waterhemp and Purlane are the four major classes in terms of image number, as opposed to weed species such as Crabgrass, Swinecress and Spurred Anoda, representing minority classes. It is clear that the present dataset has unbalanced classes. The class imbalance generally poses a challenge to machine learning modeling, which will be discussed in Sections 2.4 and 3.2.

Fig. 2 shows example images from the cotton weed dataset. The images within the same weed class have large variations in leaf color and morphology, soil background and ambient light conditions, which are desirable for building models robust to image conditions or dataset shift. The image variations vary among weed classes; despite distinct identifying characteristics, some weed classes exhibit relatively high similarities in the plant morphology. For instance, some young Morningglory and Spurred Anoda seedlings have similar, broad leaves, and the latter is also similar to Prickly Sida in terms of toothed leaf margins. Goosegrass and Crabgrass are both grassy weeds that grow prostrate on the ground, with similar leaf shapes. Palmer Amaranth and Waterhemp, which are both pigweed species, also look similar and can be difficult to distinguish from each other at early seedling stages. These similarities can contribute to errors in weed identification by DL models. A quantitative DL-based cosine similarity measure along with a similarity matrix will

be discussed in later sections (see Sections 2.5 and 3.3) to characterize the between-class similarity.

2.2. Transfer learning

Deep transfer learning (DTL) starts with a pre-trained DL model on a large-scale dataset, e.g., ImageNet and then fine-tunes the model on a new dataset from a specific domain of interest (Zhuang et al., 2020). For the weed classification in this study, we performed DTL by replacing the last fully connected (FC) layers of DL models with a layer that has 15 neurons corresponding to the same number of weed classes in the cotton weed dataset. The models were trained with by fine-tuning the weights pretrained on the ImageNet data (Russakovsky et al., 2015).

Literature review was conducted to select appropriate DL models for weed identification. The main selection criterion was the demonstrated performance of models in visual categorization tasks in the computer vision community and the availability of their source-code implementations. As a result, a suite of 35 state-of-the-art CNN models of different architectures, as summarized in Table 2, were selected for classifying the cotton weed images here. Some of them including VGG16 (Simonyan and Zisserman, 2014), ResNet50 (He et al., 2016), InceptionV3 (Szegedy et al., 2016), Xception (Chollet, 2017), and DenseNet (Huang et al., 2017), has been recently evaluated for classifying weeds in other cropping systems (Ahmad et al., 2021; Espejo-Garcia et al., 2020b; Olsen et al., 2019). The majority of these models, such as ResNeXt (Xi et al., 2017), DPN (Chen et al., 2017), EfficientNet (Tan and Le, 2019), MnasNet (Tan et al., 2019), and RepVGG (Ding et al., 2021), remain to be evaluated for weed classification tasks. For instance, RepVGG (Ding et al., 2021), which uses structural re-parameterization to decouple a training-time multi-branch topology based on VGG, was demonstrated in this study to achieve superior classification accuracies on our cotton weed dataset, and ResNeXt (Xi et al., 2017), which is built mainly upon ResNet and constructed via repeated building blocks with aggregated transformations, yielded the best classification (see Section 3).

The DL models were trained with a conventional cross entropy (CE) loss function defined as follows:

$$CE_{loss} = - \sum_{i=1}^C t_i \log(p_i) \quad (1)$$

where $p_i \in \mathbb{R}^{15}$ is a vector of a Softmax output layer (Goodfellow et al., 2016) indicating the probability of weed classes, $C = 15$ is the number of weed classes, and t_i denotes the true probability defined as:

$$t_i = \begin{cases} 1, & \text{if } i \text{ is the true label;} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

For model development and evaluation, conventionally the original dataset is randomly partitioned into training (and or validation) and test sets for model training and testing, respectively. Because of the random nature of the dataset partition, it would be desirable to run model training and testing multiple times for obtaining a reliable estimate of model performance (Raschka, 2018). Hence, in this study, instead of modeling based on dataset partition only once, we evaluated DL models using a repeated holdout validation (also called Monte Carlo cross-validation) procedure (Raschka, 2018). Specifically, DL models were trained with 5 replications, with different random seeds shared by all the models, and the mean accuracies on test data were computed for performance evaluation. In each modeling replication, the cotton weed dataset was randomly partitioned into three subsets: 65% for training, 20% for validation and 15% for testing, as shown in Fig. 1. All training and validation images were resized to 512×512 pixels in size before being fed into DL models (images of 256×512 and 1080×1080 pixels were also examined, but the size of 512×512 pixels was found to be better in terms of accuracy and speed). The image pixel intensities per color channel were normalized to the range of $[-1, 1]$ for enhanced

³ <https://forms.gle/zr9wa1uu7qHTFiK2A>.

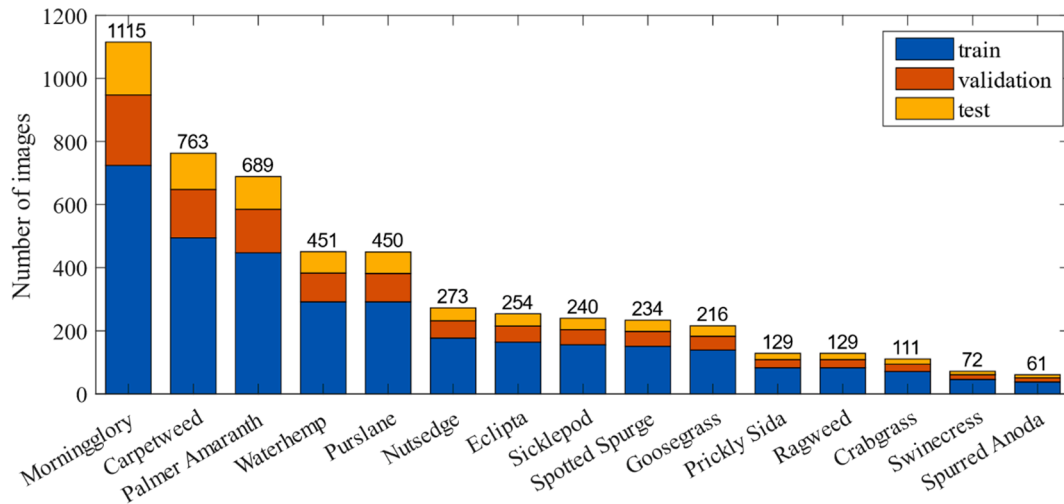


Fig. 1. Bar plot of the cotton weed dataset. Images per weed class are randomly partitioned into 65%-20%-15% splits of training, validation, and testing subsets, respectively. Numbers above the bars represent the total number of images for the corresponding weed classes.

image recognition performance (Koo and Cha, 2017). In addition, for better model accuracy, real-time data augmentation was conducted for training data by randomly rotating images in the range of $[-180^\circ, +180^\circ]$ and random flipping during the training process.

All the models were trained for 50 epochs (that were found sufficient for modeling the weed data) with the SGD (stochastic gradient descent) optimizer and a momentum of 0.9. The learning rate was initially set to be 0.001, and dynamically decreased by a factor of 0.1 every 7 epochs for stabilizing model training. The DL framework PyTorch (version 1.9) with Torchvision (version 0.10.0) (Paszke et al., 2019) was used for model training, in which a multiprocessing package⁴ was employed with 32 CPU cores to speed up the training process. The experiments were performed on an Ubuntu 20.04 server with an AMD 3990X 64-Core CPU and two GeForce RTX 3090Ti GPU (24 GB GDDR6X memory). Readers are referred to the open-source codes⁵, for a detailed implementation of transfer learning for the 35 DL models.

2.3. Performance metrics

The performance of the DL models in weed identification was evaluated in terms of the number of model parameters, training and inference times, confusion matrix and F1-score.

2.3.1. Number of model parameters

In this study, pretrained DL models were fine-tuned by updating all the model parameters for weed classification. Thus, the number of model parameters refers to all the weights (and biases) in the networks that are updated/learned during the training process through back-propagation. The parameter number is a direct measure of model complexity: networks with a larger number of parameters potentially require greater deployment memory and incur longer training and inference times (see Section 2.3.2).

2.3.2. Training and inference times

The training time is the time required to train a DL model with prescribed model configurations and computing resources. The training time depends on factors such as model architecture, number of model parameters, data size, the settings of hyper-parameters, DL framework as well as computing hardware. The training time is an important consideration where development time and resources are constrained.

A trained DL model is to be used to make predictions (also known as *inference*). The inference time (or latency time) is one crucial aspect in deploying DL models for real-time applications (e.g., in-field weed identification). It is the time that a trained DL model takes to make a prediction given an image input. In this paper, for reliable estimation, the inference time was measured as the average time needed to predict 30 weed images randomly selected from the testing dataset.

2.3.3. Confusion matrix and F1-score

The confusion matrix on testing images, which provides the accuracy for each class while revealing detailed misclassifications, was presented to show the classification for individual weed classes. The classification accuracy was measured in terms of F1 score. For the multi-class weed classification, the micro-averaged F1 score (Yang and Liu, 1999) was calculated as the classification accuracy. In micro-averaging, the per-class classifications are aggregated across classes to compute the micro-averaged precision (P) and recall (R) by counting the total true positives, false negatives, and false positives, and then a harmonic combination of P and R , i.e., Micro-F1, as follows:

$$F_1 = 2 \frac{P \times R}{P + R} \quad (3)$$

2.4. Weighted cross entropy loss

The CE loss function defined in Eq. (1) does not account for the class imbalance encountered in the cotton weed dataset (Fig. 1) in this study. Training with the CE loss may result in large classification errors for minority weed classes (e.g., Spurred Anoda). To mitigate this issue, a weighted cross entropy (WCE) (Phan and Yamamoto, 2020) loss function was introduced, performing re-weighting according to image numbers for each weed class as follows:

$$WCE_{loss} = - \sum_{i=1}^C w_i t_i \log(p_i) \quad (4)$$

where w_i is a weighting vector that assigns individualized penalty to each class, preferentially placing larger weights on minority classes. The conventional CE loss without considering class imbalance corresponds to a special WCE with a weighting vector of ones (the CE column in Table 1). In this study, an inverse-proportion weighting strategy (Phan and Yamamoto, 2020) was adopted to assign the weight to the i th weed class as follows:

⁴ <https://docs.python.org/3/library/multiprocessing.html>.

⁵ <https://github.com/Derekabc/CottonWeeds>.

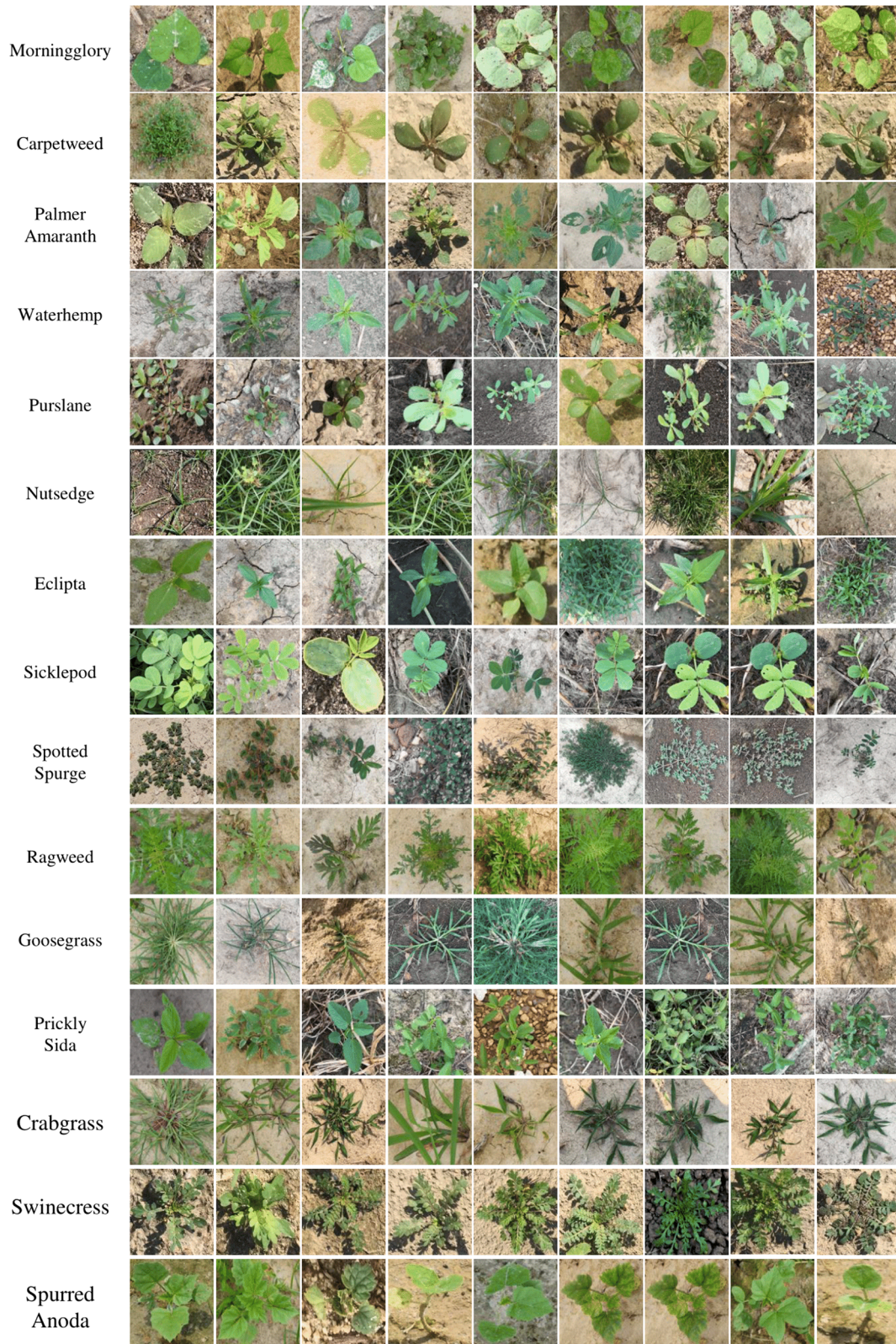


Fig. 2. Example images from the cotton weed dataset. Each row displays randomly selected images from each of the 15 weed classes.

$$w_i = \frac{N_{\max}}{N_i} \quad (5)$$

where N_i denotes the number of images for the i th weed class and $N_{\max}$ represents the maximum number of images among weed classes, i.e., 1115 (for Morningglory). As a result, weed classes with fewer images are

assigned with relatively greater weights. For example, the weight for Spurred Anoda is set to be 18.3 (1115/61). This strategy, which enforces larger penalties on misclassifications for minority classes, can potentially enhance the classification accuracy for these classes. In preliminary testing, it is observed that the direct inversion of image ratios

Table 1

Weighting coefficients of weed classes for regular CE (cross entropy) loss, Eq. (5) and the proposed WCE (weighted cross-entropy) loss.

	# of images	Weighting Coefficients		
		CE	Eq. (5)	WCE
Morningglory	1115	1.0	1.0	1.4
Carpetweed	763	1.0	1.46	2.05
Palmer Amaranth	689	1.0	1.62	2.27
Waterhemp	451	1.0	2.47	3.46
Purslane	450	1.0	2.48	3.47
Nutsedge	273	1.0	4.09	5.72
Eclipta	254	1.0	4.39	4.39
Spotted Spurge	234	1.0	4.77	4.77
Sicklepod	240	1.0	4.65	4.65
Goosegrass	216	1.0	5.16	5.16
Prickly Sida	129	1.0	8.64	8.64
Ragweed	129	1.0	8.64	8.64
Crabgrass	111	1.0	10.05	10.05
Swinecress	72	1.0	15.49	15.49
Spurred Anoda	61	1.0	18.28	18.28

may lead to sub-optimal performance, hence the final adopted weights are fine-tuned and empirically set as shown in the WCE column in Table 1. Other different choices of weighting strategies are discussed in Section 4.1.

2.5. Deep Learning-based similarity measure

To assist in the interpretation of DL classifications, an inter-class (or between-class) analysis was conducted by quantifying the similarity of the images of weed classes. Euclidean distance is the most commonly used measure of inter-class similarity, but it is sensitive to varying image conditions (e.g., ambient light, variations in camera view angle and position), which are typical of the cotton weed images collected under natural field conditions. Cosine similarity (CS), which measures the cosine of the angle between two vectors and is thus not sensitive to magnitude, offers an effective alternative to the Euclidean distance (Xia et al., 2015).

In this study, we employed a DL-based CS measure for quantifying inter-class similarities. A DL model was used as a feature extractor to obtain hierarchically learnt high-level representation of weed images, based on which the CS was calculated between two weed classes. Specifically, the VGG11 (Simonyan and Zisserman, 2014) model was trained on the cotton weed dataset through DTL, and the output of the first FC layer was taken as the feature vector, which is of length 4,096, that is, the output size of the FC layer in the VGG11 network (Simonyan and Zisserman, 2014). While other DL models can also be used for feature extraction, the VGG11 was chosen here because it achieved the best trade-off between classification performance and training time (see Table 2), particularly with high accuracies for minority weed classes (see Table 3). Given the extracted features for any two weed classes, the CS was calculated as follows (Xia et al., 2015):

$$CS(x, y) = \frac{1}{N} \sum_i \frac{x_i^T y_i}{\|x_i\| \|y_i\|} \quad (6)$$

where x_i and y_i are two feature vectors extracted by VGG11, and we randomly sample $N = 30$ pairs of images from two weed classes of interest and compute the averaged similarity value. The CS values range from -1 to 1 , where 1 means the two classes are perfectly similar and -1 means perfectly dissimilarity.

3. Experimental results

3.1. Deep learning model performance

Table 2 summarizes the number of model parameters, training, and

inference times and F1 score of the selected 35 DL models. There is a large variation in the number of model parameters, ranging from 0.74 M (million) in the SqueezeNet to 139.6 M in the VGG19. Depending on model architectures, the training time ranged from 37 mins to 168 mins. Models with a larger number of parameters tended to require a longer training time (see Fig. 3 left), because of increased model complexity. The inference times also exhibited an increasing trend with the number of parameters (see Fig. 3 right), although it had a notably smaller difference among models, ranging from 188 ms to 338 ms. Inference is mainly a forward propagation process requiring no parameter estimation and is thus far more efficient than training. Particularly, models including AlexNet, SqueezeNet, GoogleNet, ResNet18, ResNet50, VGGs, and MobileNets required inference times less than 200 ms, translating into a prediction speed of over 5 frames per second. The DL models overall show good potential to be deployed for real-time weed identification.

Fig. 4 shows the training accuracy and loss curves of the DL models. All the DL models exhibited promising training performance in terms of fast convergence speed, low training losses and high training accuracies (F1 scores). The training accuracies tended to plateau after 10 epochs at a level exceeding 90%. Regarding test accuracies (Table 2), ResNeXt101 achieved the best overall F1-score of $98.93 \pm 0.34\%$, followed by RepVGG-B1 with the $F1 = 98.90 \pm 0.34\%$. There were seven other models that yielded F1 scores exceeding 98%, including ResNeXt50, five other variants of RepVGG and DPN68. The top-10 models, as highlighted in Table 2, achieved an average F1 score of $98.54 \pm 0.30\%$. DPN68 that combines the advantages of ResNet and Densenet architectures for learning better feature representations, has demonstrated superior accuracy in visual recognition tasks (Chen et al., 2017). Despite a plain feed-forward VGG-like topology (Ding et al., 2021), RepVGG performed better such multi-branch models as ResNet and even better than the state-of-the-art models like EfficientNet. Overall the accuracies obtained by these high-performant models are higher than or comparable to those reported in literature on multi-class weed identification. For instance, Olsen et al. (2019) attained 95.1% and 95.7% accuracies by Inception V3 and ResNet50 for the eight-class DeepWeeds dataset, respectively. On the other hand, however, two models, AlexNet and MnasNet, yielded the lowest F1 scores that were close to or less than 95%, although they were all superiorly efficient to train and fast to make inferences.

The confusion matrices on test data for all the DL models are available on our Github page⁶. Due to space constraints, we only show the confusion matrices for one top-performant F1-score model, ResNeXt101, and one low-performant model, MnasNet (MnasNet1.0), in Fig. 5 and Fig. 6, respectively. The ResNeXt101 yielded accuracies of 98% or higher for 12 out of the 15 weed classes, among which 5 weeds were perfectly classified, although it misclassified 4% and 8% of the images of Swinecress and Spurred Anoda, respectively. Spurred Anoda was the most challenging weed class to distinguish from others. The ResNeXt101 achieved the classification accuracy of 92% for this species, misclassifying 2% of the weed as both Morningglory and Swinecress and 4% as Prickly Sida. Similar results are also obtained for other high-performant models such as ResNet101⁶.

The MnasNet model only achieved the accuracy of 50% in the identification of Spurred Anoda, as shown in Fig. 6, misclassifying 2%, 4%, 10%, 10% and 24% of the weed as Swinecress, Carpetweeds, Morningglory, Palmer Amaranth and Prickly Sida, respectively. The poor accuracies for this weed class are presumably because of its smallest number of images available in the dataset (61 as shown in Fig. 1). Relative low accuracies were also observed for other minority weed classes such as Goosegrass, Crabgrass and Swinecress, with accuracies of 88%, 92% and 93%, respectively, by the MasNet. To improve

⁶ https://github.com/Derekabc/CottonWeeds/tree/master/Confusing_Matrices.

Table 2

Performance of 35 state-of-the-art deep learning models with five random holdout validations on the cotton weed dataset. Note that since the variations in training time are negligible, its standard deviation is not included. The top-10 testing accuracies of DL models are highlighted in bold. "M" stands for a million.

Index	Models		Parameter Number	Training Time	Training F1-Score	Testing F1-Score	Inference Time (ms)
1	AlexNet		57.1 M	37 m 2 s	95.53 ± 0.21	94.48 ± 0.53	188.5 ± 2.2
2	SqueezeNet		0.74 M	46 m 7 s	96.14 ± 0.46	95.17 ± 0.73	187.3 ± 1.6
3	GoogleNet		5.6 M	52 m 28 s	94.71 ± 0.23	96.70 ± 0.42	196.3 ± 0.5
4	Xception		20.8 M	89 m 9 s	94.87 ± 0.29	96.16 ± 0.55	211.3 ± 1.8
5	DPN68		11.8 M	79 m 10 s	98.67 ± 0.11	98.02 ± 0.11	219.0 ± 6.9
6	MnasNet		3.1 M	51 m 3 s	91.95 ± 0.30	94.78 ± 1.06	191.2 ± 2.0
7	ResNet	ResNet18	11.2 M	47 m 30 s	96.92 ± 0.14	97.14 ± 0.46	188.9 ± 0.9
8		ResNet50	23.5 M	73 m 17 s	98.14 ± 0.14	97.91 ± 0.34	195.6 ± 0.4
9		ResNet101	42.5 M	92 m 55 s	98.44 ± 0.14	97.97 ± 0.16	207.0 ± 0.6
10	ResNeXt	ResNeXt50	23.0 M	63 m 10 s	98.56 ± 0.13	98.80 ± 0.48	322.4 ± 0.3
11		ResNeXt101	86.8 M	168 m 17 s	98.90 ± 0.12	98.93 ± 0.34	338.5 ± 0.1
12	VGG	VGG11	128.8 M	67 m 46 s	97.56 ± 0.29	96.49 ± 0.63	194.1 ± 1.3
13		VGG16	134.3 M	99 m 25 s	97.93 ± 0.17	96.87 ± 0.45	195.7 ± 1.4
14		VGG19	139.6 M	112 m 41 s	97.91 ± 0.11	97.01 ± 0.57	197.2 ± 1.4
15	Densenet	Densenet121	7.0 M	75 m 40 s	98.02 ± 0.13	97.79 ± 0.32	212.4 ± 0.8
16		Densenet161	26.5 M	133 m 42 s	98.48 ± 0.16	97.85 ± 0.27	227.4 ± 0.5
17		Densenet169	12.5 M	85 m 1 s	98.24 ± 0.16	97.85 ± 0.27	226.8 ± 0.5
18	Inception	Inception V3	24.4 M	73 m 50 s	96.78 ± 0.17	97.47 ± 0.11	206.3 ± 0.4
19		Inception V4	41.2 M	120 m 42 s	95.88 ± 0.20	96.93 ± 0.15	235.4 ± 0.8
20		Inception-ResNet V2	54.3 M	124 m 36 s	93.78 ± 0.04	96.16 ± 0.21	255.9 ± 1.4
21	Mobilenet	Mobilenet V2	2.2 M	53 m 27 s	97.39 ± 0.14	97.41 ± 0.35	191.1 ± 0.8
22		MobilenetV3-small	1.5 M	41 m 27 s	94.27 ± 0.35	95.72 ± 0.82	193.1 ± 1.2
23		MobilenetV3-large	4.2 M	49 m 4 s	96.69 ± 0.19	96.89 ± 0.54	193.8 ± 2.0
24	EfficientNet	EfficientNet-b0	4.0 M	63 m 39 s	93.10 ± 0.32	95.95 ± 0.91	202.0 ± 5.6
25		EfficientNet-b1	6.5 M	77 m 8 s	93.74 ± 0.18	95.80 ± 0.70	203.8 ± 0.8
26		EfficientNet-b2	7.7 M	78 m 56 s	94.16 ± 0.22	96.36 ± 0.58	204.5 ± 1.7
27		EfficientNet-b3	10.7 M	92 m 51 s	95.11 ± 0.33	96.74 ± 0.47	211.3 ± 1.2
28		EfficientNet-b4	17.6 M	113 m 12 s	94.23 ± 0.69	95.74 ± 0.69	216.3 ± 1.3
29		EfficientNet-b5	28.4 M	144 m 44 s	94.07 ± 0.27	95.57 ± 0.53	224.1 ± 1.5
30	RepVGG	RepVGG-A0	7.8 M	32 m 14 s	97.61 ± 0.17	98.24 ± 0.39	320.9 ± 0.2
31		RepVGG-A1	12.8 M	31 m 16 s	97.96 ± 0.18	98.35 ± 0.31	322.3 ± 0.2
32		RepVGG-A2	26.8 M	44 m 17 s	98.17 ± 0.20	98.70 ± 0.32	323.8 ± 0.2
33		RepVGG-B0	14.6 M	37 m 35 s	98.07 ± 0.17	98.65 ± 0.33	327.4 ± 0
34		RepVGG-B1	55.4 M	75 m 14 s	98.45 ± 0.10	98.90 ± 0.34	330.3 ± 0.1
35		RepVGG-B2	86.5 M	112 m 23 s	98.53 ± 0.12	98.88 ± 0.26	333.1 ± 0

Table 3

Performance comparison between CE (cross entropy) loss and WCE (weighted cross entropy) loss for the top and bottom four weed classes and the overall F1-score (%).

Model		GoogleNet	Densenet	Xception	Mnasnet	EfficientNet	VGG11	RepVGG
Morgingglory	CE	99.05	98.81	98.57	97.14	98.10	97.62	99.29
	WCE	97.26	98.93	96.55	93.0.81	95.83	96.55	97.98
Waterhemp	CE	98.53	99.12	99.71	98.82	99.12	98.24	99.41
	WCE	98.53	99.41	99.12	98.24	99.12	97.65	99.71
Palmer Amaranth	CE	94.81	97.88	94.42	93.65	95.19	95.00	96.92
	WCE	93.08	96.92	93.65	90.58	93.85	94.42	96.15
Purslane	CE	97.94	98.82	99.71	95.59	97.94	97.06	98.53
	WCE	97.35	99.71	97.94	95.00	98.53	96.76	98.24
Ragweed	CE	100	100	99.00	96.00	100	98.00	100
	WCE	100	100	99.00	100	100	99.00	100
Crabgrass	CE	94.12	97.65	89.41	91.76	90.59	96.47	97.65
	WCE	100	100	98.82	97.65	100	97.65	97.65
Swinecress	CE	92.73	96.36	90.91	92.73	92.73	96.36	96.36
	WCE	96.36	96.36	96.36	94.55	96.36	96.36	96.36
Spurred Anoda	CE	56.00	90.00	48.00	50.00	62.00	76.00	86.00
	WCE	86.00	92.00	90.00	82.00	90.00	88.00	90.00
Overall F1score (%)	CE	97.80	97.85	96.16	94.78	96.36	96.49	98.65
	WCE	97.07	97.74	96.59	94.92	96.07	96.47	98.27

Note: because of space constraints, Densenet, Efficient and RepVGG represent Densenet161, EfficientNet-b2 and RepVGG-B0, respectively.

the performance of DL models in classifying the minority weed classes, the proposed WCE loss function is discussed next.

3.2. Performance improvement with the WCE loss

Fig. 7 shows the confusion matrix achieved by the MnasNet model trained with the WCE loss function as given in Eq. (4). The WCE-based model achieved remarkable improvements over the counterpart

(Fig. 6) trained with the regular CE function in Eq. (1) in classifying minority weed classes. The classification accuracy of Spurred Annoda jumped from 50% to 82%. The accuracies for other minority weed classes including Goosegrass, Crabgrass and Swinecress were improved from 88% to 93%, 92% to 98% and 93% to 95%, respectively.

Table 3 compares the classification accuracies by seven selected models (including the aforementioned MnasNet) using the CE loss and the WCE loss. The confusion matrices for all the DL models trained with

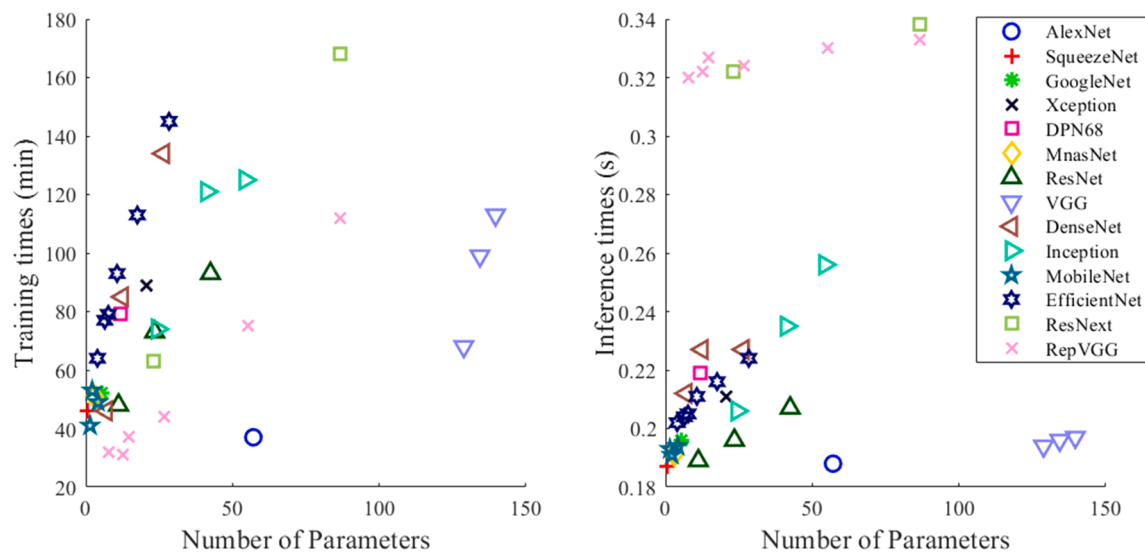


Fig. 3. Training and inference time v.s. number of parameters. The deep learning models from the same family are labeled with the same marker.

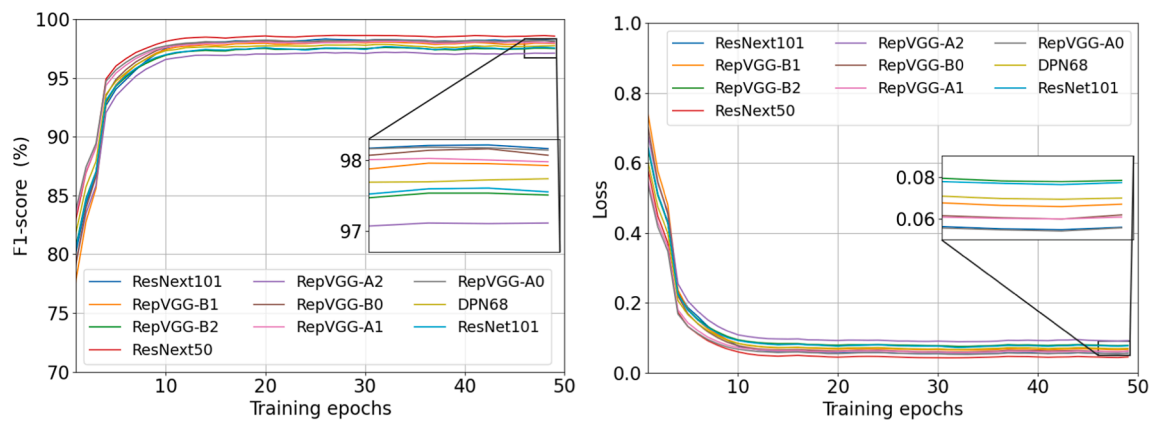


Fig. 4. Training accuracy and loss curves for the deep learning models on the training dataset.

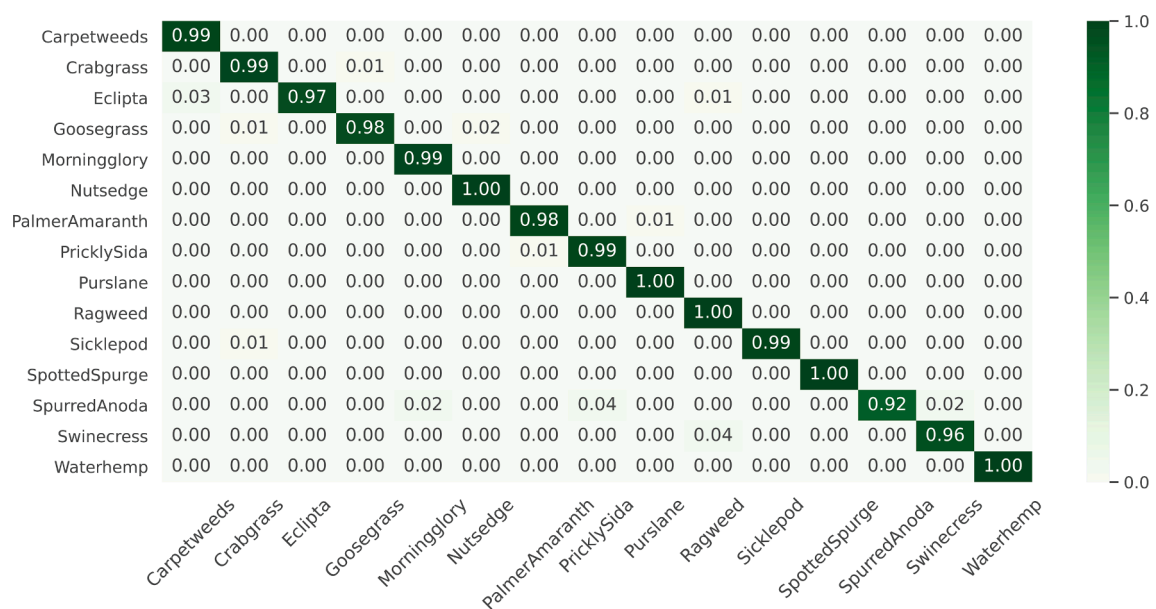


Fig. 5. The confusion matrix of the ResNeXt101 model on the test dataset.

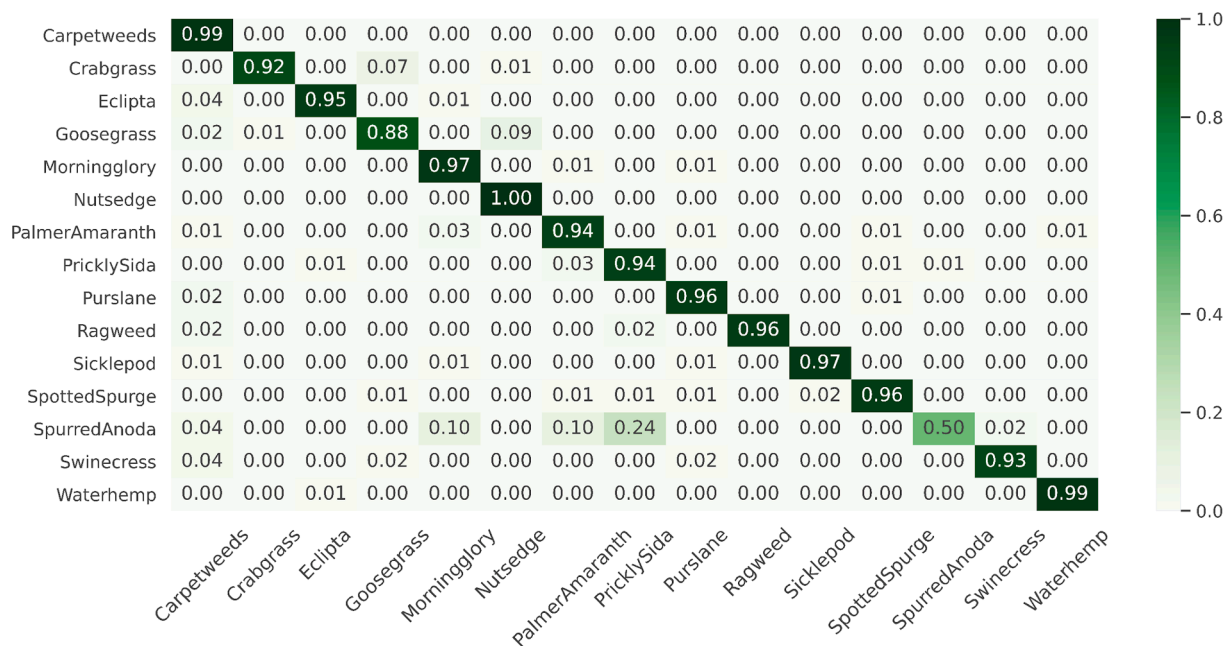


Fig. 6. The confusion matrix of the MnasNet model on the test dataset.

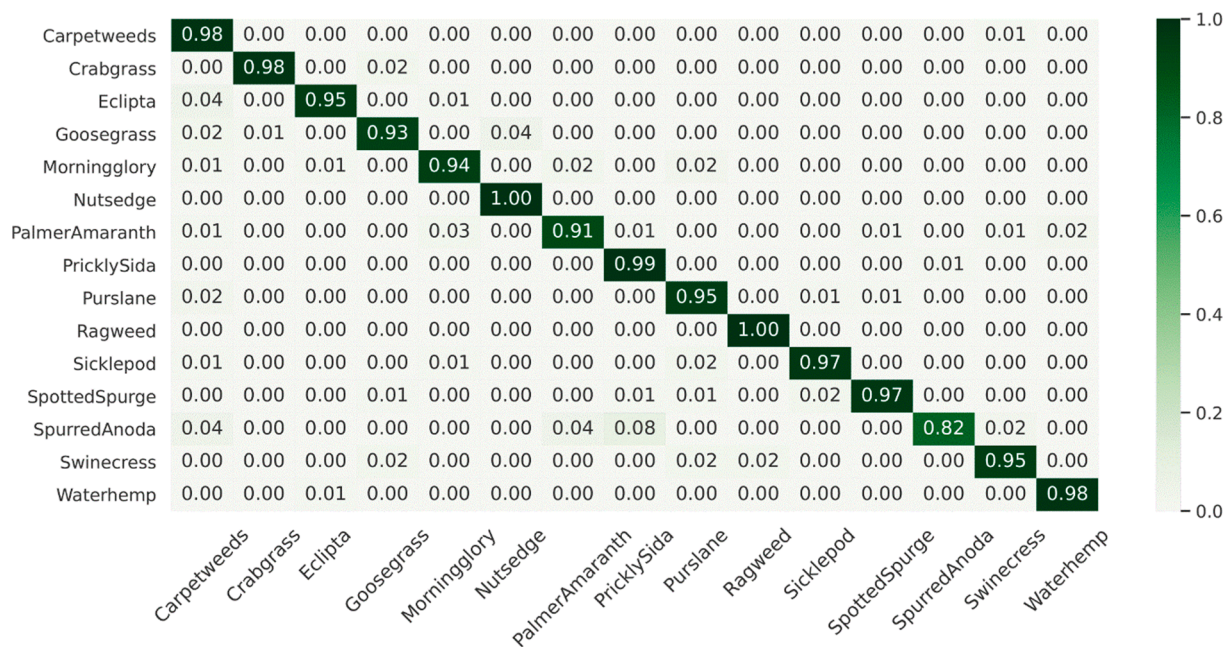


Fig. 7. The confusion matrix of the MnasNet model with the WCE (weighted cross entropy) loss strategy evaluated on the test dataset.

the CE and WCE loss functions separately are available at the site⁷. Emphasis here is placed on the classification results for the four minority weeds including Ragweed, Crabgrass, Swinecress and Spurred Anoda, alongside the four majority weeds including Morningglory, Waterhemp, Palmer Amaranth and Purslane. Considerable improvements were achieved by all these models for the minority weed classes. Notably, in addition to MnasNet that resulted in an improvement of 32% in identifying Spurred Anoda, Xception, GoogleNet and EfficientNet-b2 also achieved dramatic improvements of 42%, 30% and 28% in identifying

this species, respectively, compared to the models with the CE loss.

Despite improvements on minority classes, models like Xception and MnasNet resulted in a slightly decreased accuracy for Morningglory. This is because the WCE strategy that placed stronger weights on minority classes might negatively affect the classification for major classes. Nonetheless, the significant improvements on the minority classes outweighed the decrease accuracy in the majority classes, leading to overall improvements in F1-score by these models.

3.3. Weed similarity analysis

Fig. 8 shows an inter-class CS (cosine similarity) matrix based on the features extracted by the VGG11 model (with the CE loss) (Section 2.5).

⁷ https://github.com/Derekabc/CottonWeeds/tree/master/Confusing_Matrices.

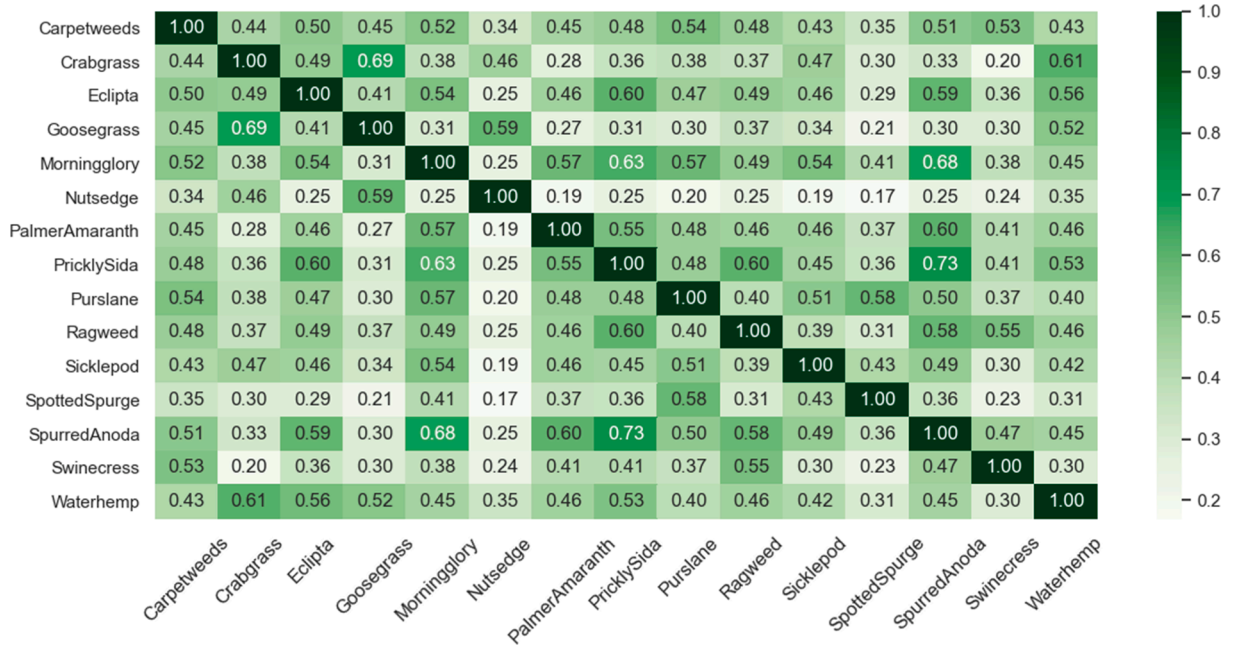


Fig. 8. The similarity matrix achieved by the deep learning-based cosine similarity measure (Section 2.5) on the cotton weed dataset. The diagonal entries indicate the perfect similarity of each class with itself.

The CS matrix helps explain misclassifications by DL models among weed classes. Weed classes that share more common features tend to have higher CS values. For example, Goosegrass and Crabgrass that are both grassy weeds in the Poaceae family, had a CS of 0.69, greater than their similarities with all other weeds, and Goosegrass also shared high similarity with Nutsedge with a CS of 0.59. The high CS between two classes agrees well with the associated classification errors (see Fig. 5, Fig. 6 and Fig. 7). For instance, for the MnasNet model (Fig. 6), 7% misclassifications for Crabgrass were due to misclassifying the weed as Goosegrass and 9% errors for Goosegrass were due to misclassifications into Nutsedge. Spurred Anoda and Prickly Sida are another pair of similar weeds, which both have toothed leaf margins and are members of the Mallow family. The globally highest CS of 0.73 was observed between the two classes. Their strong similarity, along with the fact that Prickly Sida has more than twice as many images as Spurred Anoda, explains the significant proportion of Spurred Anoda misclassified as Prickly Sida (see Fig. 5, Fig. 6, and Fig. 7).

4. Discussion and future research

In this section, we discuss two potential approaches to improving the performance on minority weed classes, which will be investigated in future studies.

4.1. Weighted loss functions

The WCE loss function in Eq. (4) improves the CE loss by adaptively assigning weights to individual weed classes to account for class imbalance. In addition to the weighting in Eq. (5), there are other weighting (Phan and Yamamoto, 2020) cost-sensitive methods (Khan et al., 2017) to cope with imbalanced data. The class-balanced (CB) loss introduced in (Cui et al., 2019) re-balances the classification loss based on the effective number of samples for each class.

The CB loss is defined as:

$$CB_{loss} = - \sum_{i=1}^C \frac{1-\beta}{1-\beta^{\alpha_i}} t_i \log(p_i) \quad (7)$$

when $\beta = 0$ the CB loss is equivalent to the CE loss and $\beta \rightarrow 0$ corresponds

to re-weighting by inverse class frequency, which enables smoothly adjusting the class-balanced term between no re-weighting and re-weighting by inverse class frequency (Cui et al., 2019).

Focal loss (FL), which was originally proposed in Lin et al. (2017), offers another promising alternative for imbalanced learning, which is calculated as follows:

$$FL_{loss} = \sum_{i=1}^C (1-p_i)^{\gamma} t_i \log(p_i) \quad (8)$$

where $(1-p_i)^{\gamma}$ is called the modulating factor, which allows down weighting the contributions of easy examples or majority classes during training while rapidly focusing on challenging classes that have few images. Here, $\gamma \geq 0$ is the focusing parameter, and FL loss is reduced to the conventional CE loss when equivalent to $\gamma = 0$. In future research, we will experiment and evaluate these weighted loss functions for improved classification of minority weed classes.

4.2. Data augmentation

In this paper, although overall DL models achieved remarkable weed identification accuracy, some models that prove to be powerful in visual categorization tasks, like EfficientNet (Tan and Le, 2019), did not perform well as expected, especially on minority weed classes⁸. This is likely due to the fact that these models are heavily reliant on large-scale data to be sufficiently optimized while avoiding overfitting (Shorten and Khoshgoftaar, 2019). One intuitive solution is to collect more images for the under-performed weed classes. However, it may be difficult to collect new images for many weed species in practice due to unpredicted weather conditions and limited access to a diversity of field sites.

Data augmentation (DA) offers a potentially effective means to address the insufficiency of physically collected image data. In DA, a suite of techniques (Shorten and Khoshgoftaar, 2019), such as geometric transformations, color space augmentations and especially generative adversarial networks (GANs) (Olaniyi et al., 2022), can be used to

⁸ https://github.com/Derekabc/CottonWeeds/tree/master/Confusing_Matrices.csv.

enhances the size and quality of training images, such that deep learning models can be trained on the artificially expanded dataset and gain better performance. Particularly, GANs, representing a novel framework of generative modeling through adversarial training, have received increasing attention in DL for agriculture (Olaniyi et al., 2022). Recently, GAN methods have been investigated for weed recognition tasks (Espejo-Garcia et al., 2021; Fawkerherji et al., 2021) to address the lack of large-scale domain datasets. In this paper, although geometric transformation methods, including random flipping and rotation and pixel normalization, were employed for image augmentation, they were still limited in terms of image variations generated and did not fully showcase the potential of DA techniques for DL-based weed classification, which will be a subject of future research.

5. Conclusion

In this study, a first, comprehensive benchmark of a suite of 35 DL models was established through transfer learning for multi-class identification of common weeds specific to cotton production systems. A dedicated dataset consisting of 5187 images of 15 weed classes was created by collecting images under natural light conditions and at varied weed growth stages from a diversity of cotton fields in southern U.S. states over two growth seasons. DTL proved to be effective for achieving high weed classification accuracies (F1 score > 95%) within reasonably short training time (<2.5 h). ResNeXt was found to be the best-performant model in terms of the highest overall accuracy of $F1 = 98.93 \pm 0.34\%$, and the top-10 models resulted in an average F1-score of $98.54 \pm 0.30\%$. A WCE loss function was proposed for model training, in which individualized weights were assigned to weed classes to account for class imbalance, which achieved substantial improvements in classifying minority weed classes. A DL-based cosine similarity metric was found to be useful for assisting the interpretation of the misclassifications. Both the source codes for model development and evaluation and the cotton weed dataset were made publicly accessible for the research community. This study lays the foundation for informed choice of DL models for weed classification tasks and can be beneficial for precision agriculture research at large.

CRedit authorship contribution statement

Dong Chen: Formal analysis, Software, Writing – original draft. **Yuzhen Lu:** Conceptualization, Investigation, Data curation, Supervision, Writing – original draft, Writing – review & editing. **Zhaojian Li:** Resources, Writing – review & editing. **Sierra Young:** Data curation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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